LESSON 08 · MISSION GUIDE

A Thousand Arrows at Once

NumPy: do the maths on 400 quests in one line, no loop, no dragging.

From **Orrin of the Clockworks**
To **Jen, Keeper of the Guild Ledger**
By **raven (the third one this week about the loop)**

Keeper. I watched the Grimoire do the quest tally at the last full moon. It walks the ledger one quest at a time, recalculating the whole workbook between each. Forty seconds. Sometimes it skips the first row; nobody knows why. The Quartermistress brews tea while it runs. She has a dedicated mug. Biscuit has started sitting on it.

NumPy treats a whole column as one thing: `paid / days` divides all 400 at once. A legion of archers loosing together, not a queue. It's dragging a formula down, except it's instant and never skips the first row. Install it with pip, then go and retire the Quartermistress's mug. — O.

What you'll be able to do after today

- [ ] Turn a list of numbers into a NumPy **array** and ask it for min, max, mean, median, sum
- [ ] Do **column maths** with no loop: `paid / days` gives 400 crowns-per-quest-day at once
- [ ] Filter with **boolean masks**: `rewards[rewards > 1000]` is AutoFilter in one line
- [ ] Combine filters with `&`, `|` and `~`
- [ ] Compute **percentiles and quartiles**, the ones the Royal Auditors ask for
- [ ] Work with a 2-D array (a whole range) and sum it along either axis

Why this matters at work

- **This is the Charter question.** Exercise 4 shows, from all 400 quests, how success falls as difficulty rises. Exercise 5 finds which quests are worth the crowns. Those are two of the three things the Queen wants to hear at Midsummer.
- **pandas is built on NumPy.** Next lesson's DataFrame columns *are* arrays. Everything you learn today (masks, axis, vectorised maths) carries over directly.

Four big ideas

1. An array is a column	`np.array([1, 2, 3])`. One type for the whole column (`dtype`), one shape. Like a column of numbers in Excel, not a list of mixed things.
2. Maths is element-wise	`rewards * 1.10` raises every reward. `paid / days` divides row by row. No loop. That's "vectorised".
3. A comparison gives a mask	`rewards > 1000` is an array of True/False. Put it in square brackets to filter: `rewards[mask]`. `mask.sum()` counts the Trues.
4. Axis	In a 2-D array, `axis=0` works down the columns, `axis=1` along the rows. SUM at the bottom of a column vs SUM at the end of a row.

Three things that will bite you

- `and` / `or` don't work on arrays. Use `&` / `|`, and put brackets round each comparison.
- `np.percentile(x, 75)`, not `0.75`. Excel's `PERCENTILE` wants 0.75; NumPy wants 75.
- `rewards.mean` without brackets is the method, not the number. If you see `<built-in method mean>`, add `()`.

How today works

15 MIN
Lecture
Lars does the quest tally in one line, then explains it

60 MIN
Exercise sheet
Seven ravens about the 400 quests in the guild ledger

15 MIN
Pages from the Grimoire, page 8
Dalvric's danger report. The Queen still has the error

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Pell	Your first column	`np.array()` `.shape` `.min()` `.mean()` `np.median()`
2	Pell	Dragging a formula down	array maths, `.round(2)`
3	Siv	AutoFilter, but faster	masks, `&` `|` `~`, `.sum()`
4	The Queen	Too dangerous?	a mask per difficulty, mean vs median
5	Pell	Lean quests and rich quests	`np.percentile()`
6	Orrin	A whole range at once	2-D arrays, `axis=`, `[row, col]`
7 ⭐	The Queen	What if?	`np.where()`, `np.maximum()`, a function

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if the Privy Council can wait), each printing the expected output
- [ ] Dalvric's danger report runs and prints the expected three lines, with one commit per bug
- [ ] Everything is pushed to `lesson-08` and you've opened a pull request



Next lesson: The Living Ledger. pandas gives you a DataFrame: a whole table with named columns, filters, sorting and reading straight from Excel. The guild's old ledger workbook, come to life.